

The Addison Wesley Signature Series

A MIKE COHN SIGNATURE
BOOK
Mike Cohn

MORE AGILE TESTING

LEARNING JOURNEYS FOR
THE WHOLE TEAM

JANET GREGORY
LISA CRISPIN



Forewords by Elisabeth Hendrickson and Johanna Rothman

Praise for *More Agile Testing*

“I love this book. It will help to create really great testers. That’s a good thing, since anyone who reads this will want to have one on their team.”

—*Liz Keogh, agile coach, Lunivore Limited*

“This book will change your thinking and move your focus from *tests* to *testing*. Yes, it is not about the result, but about the activity!”

—*Kenji Hiranabe, cofounder of Astah and CEO, Change Vision, Inc.*

“To my mind, agile development is about learning—that one word captures the true spirit of what agile is all about. When I had the chance to read through their new book, I could only say, ‘Wow! Janet and Lisa have done themselves proud.’ This is not a book about testing; this is a book about learning. Their clear explanations are accompanied by great true stories and an impressive list of books, articles, and other resources. Those of us who like learning, who love to dig for more information, can rejoice! I know you’re always looking for something interesting and useful; I can guarantee that you will find it here!”

—*Linda Rising, coauthor of Fearless Change: Patterns for Introducing New Ideas*

“Janet and Lisa’s first book, *Agile Testing*, drew some general principles that are still important today but left me wondering, ‘how?’ In this second book, they adapt those principles to today’s development landscape—with mobile, DevOps, and cloud-based applications delivered in increasingly compressed release cycles. Readers get specific testing tools for the mind along with new practices and commentary to accelerate learning. Read it today.”

—*Matt Heusser, Managing Principal, Excelon Development*

“An excellent guide for your team’s agile journey, full of resources to help you with every kind of testing challenge you might meet along the way. Janet and Lisa share a wealth of experience with personal stories about how they helped agile teams figure out how to get value from testing. I really like how the book is filled with techniques explained by leading industry practitioners who’ve pioneered them in their own organizations.”

—*Rachel Davies, agile coach, unruly and coauthor of Agile Coaching*

“Let me net this out for you: agile quality and testing is hard to get right. It’s nuanced, context-based, and not for the faint of heart. In order to effectively balance it, you need hard-earned, pragmatic, real-world advice. This book has it—not only from Janet and Lisa, but also from forty additional expert agile practitioners. Get it and learn how to effectively drive quality into your agile products and across your entire organization.”

—*Bob Galen, Principal Consultant, R Galen Consulting Group, and Author of Agile Reflections and Scrum Product Ownership*

“Janet and Lisa have done it again. They’ve combined pragmatic life experience with ample storytelling to help people take their agile testing to the next level.”

—Jonathan Rasmusson, *author of Agile Samurai: How Masters Deliver Great Software*

“In this sequel to their excellent first book, Janet and Lisa have embraced the maturity of agile adoption and the variety of domains in which agile approaches are now being applied. In *More Agile Testing* they have distilled the experiences of experts working in different agile organizations and combined them with their own insights into a set of invaluable lessons for agile practitioners. Structured around a range of essential areas for software professionals to consider, the book examines what we have learned about applying agile, as its popularity has grown, and about software testing in the process. There is something for everyone here, not only software testers, but individuals in any business role or domain with an interest in delivering quality in an agile context.”

—Adam Knight, *Director of QA, RainStor*

“This book has it all: practical advice and stories from the trenches. Whether you’ve never heard of agile or you think you’re an expert, there is something here that will help you out. Jump around in the book and try a few things; I promise you will be a better tester and developer for it.”

—Samantha Laing, *agile coach and trainer, Growing Agile*

“*More Agile Testing* is a great collection of stories and ideas that can help you improve not just how you test, but the products you build and the way you work. What I love most about the book is how easy many of the ideas are to try. If one message is clear, it is that regardless of your context and challenges, there are things you can try to improve. Get started today with something small, and nothing will be impossible.”

—Karen Greaves, *agile coach and trainer, Growing Agile*

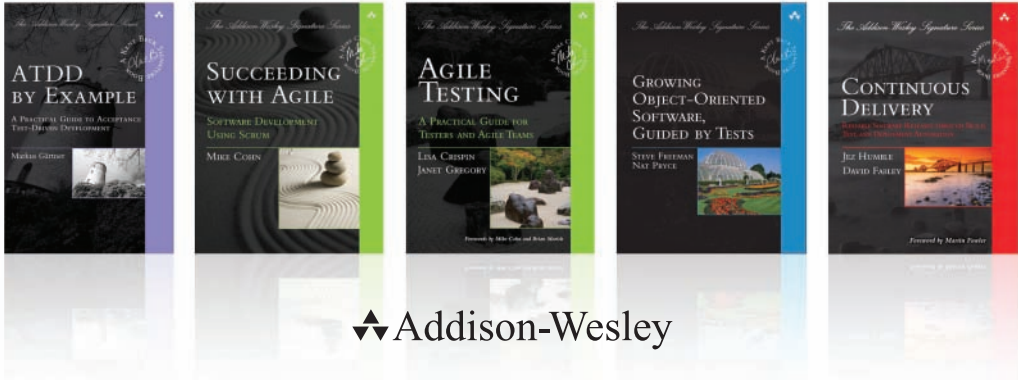
“*More Agile Testing* is an extensive compilation of experiences, stories, and examples from practitioners who work with testing in agile environments around the world. It covers a broad spectrum, from organizational and hiring challenges, test techniques and practices, to automation guidance. The diversity of the content makes it a great cookbook for anyone in software development who is passionate about improving their work and wants to produce quality software.”

—Sigurdur Birgisson, *quality assistance engineer, Atlassian*

MORE AGILE TESTING

The Addison-Wesley Signature Series

Kent Beck, Mike Cohn, and Martin Fowler, Consulting Editors



Visit informit.com/awss for a complete list of available products.

The Addison-Wesley Signature Series provides readers with practical and authoritative information on the latest trends in modern technology for computer professionals. The series is based on one simple premise: Great books come from great authors. Titles in the series are personally chosen by expert advisors, world-class authors in their own right. These experts are proud to put their signatures on the covers, and their signatures ensure that these thought leaders have worked closely with authors to define topic coverage, book scope, critical content, and overall uniqueness. The expert signatures also symbolize a promise to our readers: You are reading a future classic.



Make sure to connect with us!
informit.com/socialconnect



informIT.com
THE TRUSTED TECHNOLOGY LEARNING SOURCE

Safari
Books Online

MORE AGILE TESTING

LEARNING JOURNEYS FOR THE WHOLE TEAM

Janet Gregory
Lisa Crispin

◆ **Addison-Wesley**

Upper Saddle River, NJ • Boston • Indianapolis • San Francisco
New York • Toronto • Montreal • London • Munich • Paris • Madrid
Capetown • Sydney • Tokyo • Singapore • Mexico City

Many of the designations used by manufacturers and sellers to distinguish their products are claimed as trademarks. Where those designations appear in this book, and the publisher was aware of a trademark claim, the designations have been printed with initial capital letters or in all capitals.

The authors and publisher have taken care in the preparation of this book, but make no expressed or implied warranty of any kind and assume no responsibility for errors or omissions. No liability is assumed for incidental or consequential damages in connection with or arising out of the use of the information or programs contained herein.

For information about buying this title in bulk quantities, or for special sales opportunities (which may include electronic versions; custom cover designs; and content particular to your business, training goals, marketing focus, or branding interests), please contact our corporate sales department at corpsales@pearsoned.com or (800) 382-3419.

For government sales inquiries, please contact governmentsales@pearsoned.com.

For questions about sales outside the United States, please contact international@pearsoned.com.

Visit us on the Web: informit.com/aw

Library of Congress Cataloging-in-Publication Data

Gregory, Janet, 1953–

More agile testing : learning journeys for the whole team / Janet Gregory, Lisa Crispin.
pages cm

Includes bibliographical references and index.

ISBN 978-0-321-96705-3 (pbk. : alk. paper)

1. Computer software—Testing. 2. Agile software development. I. Crispin, Lisa. II. Title.

QA76.76.T48G74 2015

005.1—dc23

2014027150

Copyright © 2015 Janet Gregory and Lisa Crispin

Illustrations by Jennifer Sinclair

All rights reserved. Printed in the United States of America. This publication is protected by copyright, and permission must be obtained from the publisher prior to any prohibited reproduction, storage in a retrieval system, or transmission in any form or by any means, electronic, mechanical, photocopying, recording, or likewise. To obtain permission to use material from this work, please submit a written request to Pearson Education, Inc., Permissions Department, 200 Old Tappan Road, Old Tappan, New Jersey 07675, or you may fax your request to (201) 236-3290.

ISBN-13: 978-0-321-96705-3

ISBN-10: 0-321-96705-4

Text printed in the United States on recycled paper at RR Donnelley in Crawfordsville, Indiana.

Second printing, October 2015

*To my grandchildren, Lauren, Brayden, and Joe, who kept
me laughing and playing throughout this past year.*

—Janet

*To my family, those still here and those sadly gone, and my dear friends
who are part of my chosen family.*

—Lisa

This page intentionally left blank

CONTENTS

Foreword by Elisabeth Hendrickson	xvii
Foreword by Johanna Rothman	xix
Preface	xxi
Acknowledgments	xxix
About the Authors	xxxiii
About the Contributors	xxxv

Part I	Introduction	1
Chapter 1	How Agile Testing Has Evolved	3
	Summary	6
Chapter 2	The Importance of Organizational Culture	7
	Investing Time	8
	The Importance of a Learning Culture	12
	Fostering a Learning Culture	13
	Transparency and Feedback Loops	15
	Educating the Organization	17
	Managing Testers	19
	Summary	20

Part II	Learning for Better Testing	21
Chapter 3	Roles and Competencies	23
	Competencies versus Roles	24
	T-Shaped Skill Set	28
	Generalizing Specialists	33
	Hiring the Right People	36
	Onboarding Testers	37
	Summary	39
Chapter 4	Thinking Skills for Testing	41
	Facilitating	42
	Solving Problems	43
	Giving and Receiving Feedback	45
	Learning the Business Domain	46
	Coaching and Listening Skills	48
	Thinking Differently	49
	Organizing	51
	Collaborating	52
	Summary	53
Chapter 5	Technical Awareness	55
	Guiding Development with Examples	55
	Automation and Coding Skills	56
	General Technical Skills	59
	Development Environments	59
	Test Environments	60
	Continuous Integration and Source Code Control Systems	62
	Testing Quality Attributes	65
	Test Design Techniques	67
	Summary	67
Chapter 6	How to Learn	69
	Learning Styles	69
	Learning Resources	72
	Conferences, Courses, Meet-ups, and Collaborating	72
	Publications, Podcasts, and Online Communities	75
	Time for Learning	77
	Helping Others Learn	79
	Summary	83

Chapter 10 The Expanding Tester's Mindset: Is This My Job?	137
Whose Job Is This Anyway?	137
Business Analysis Skills	137
UX Design Skills	140
Documentation Skills	141
Take the Initiative	142
Summary	144
Chapter 11 Getting Examples	145
The Power of Using Examples	145
Guiding Development with Examples	148
ATDD	149
BDD	152
SBE	153
Where to Get Examples	155
Benefits of Using Examples	157
Potential Pitfalls of Using Examples	159
Getting Bugged Down in the Details	159
Lacking Buy-in	160
Too Many Regression Tests	161
Not Enough Is Known Yet	161
The Mechanics of Using Examples to Guide Coding	162
Summary	162
Part V Investigative Testing	163
Chapter 12 Exploratory Testing	165
Creating Test Charters	168
Generating Test Charter Ideas	171
Exploring with Personas	171
Exploring with Tours	174
Other Ideas	175
Managing Test Charters	176
Session-Based Test Management	176
Thread-Based Test Management	178
Exploring in Groups	183
Recording Results for Exploratory Test Sessions	185
Where Exploratory Testing Fits into Agile Testing	188
Summary	190

Chapter 13 Other Types of Testing	191
So Many Testing Needs	192
Concurrency Testing	194
Internationalization and Localization	195
Regression Testing Challenges	200
User Acceptance Testing	201
A/B Testing	203
User Experience Testing	205
Summary	207

Part VI Test Automation 209

Chapter 14 Technical Debt in Testing	211
Make It Visible	212
Work on the Biggest Problem—and Get the Whole Team Involved	217
Summary	220
Chapter 15 Pyramids of Automation	223
The Original Pyramid	223
Alternate Forms of the Pyramid	224
The Dangers of Putting Off Test Automation	227
Using the Pyramid to Show Different Dimensions	231
Summary	235
Chapter 16 Test Automation Design Patterns and Approaches	237
Involve the Whole Team	238
Starting Off Right	239
Design Principles and Patterns	240
Testing through the API (at the Service Level)	241
Testing through the UI	243
Test Maintenance	248
Summary	251
Chapter 17 Selecting Test Automation Solutions	253
Solutions for Teams in Transition	253
Meeting New Automation Challenges with the Whole Team	258
Achieving Team Consensus for Automation Solutions	260

How Much Automation Is Enough?	262
Collaborative Solutions for Choosing Tools	264
Scaling Automation to Large Organizations	264
Other Automation Considerations	268
Summary	269

Part VII What Is Your Context? 271

Chapter 18 Agile Testing in the Enterprise	275
What Do We Mean by “Enterprise”?	275
“Scaling” Agile Testing	276
Dealing with Organizational Controls	278
Coordinating Multiple Teams	283
A System Test Team and Environment	284
Consistent Tooling	289
Coordination through CI	289
Version Control Approaches	290
Test Coverage	291
Managing Dependencies	292
Working with Third Parties	292
Involving Customers in Large Organizations	294
Advantages of Reaching Out beyond the Delivery Team	296
Summary	297
Chapter 19 Agile Testing on Distributed Teams	299
Why Not Colocate?	301
Common Challenges	302
Cultural Issues	302
Language	303
Time Zones	304
Dependencies	305
Planning	305
Strategies for Coping	308
Integrating Teams	308
Communication and Collaboration	309
Collaborating through Tests	311
Offshore Testing	312

Tool Ideas for Distributed Teams	319
Communication Tools	319
Collaboration Tools	319
Summary	322
Chapter 20 Agile Testing for Mobile and Embedded Systems	325
Similar, Yet Different	326
Testing Is Critical	328
Agile Approaches	329
Summary	337
Chapter 21 Agile Testing in Regulated Environments	339
The “Lack of Documentation” Myth	339
Agile and Compliance	340
Summary	346
Chapter 22 Agile Testing for Data Warehouses and Business Intelligence Systems	347
What Is Unique about Testing BI/DW?	348
Using Agile Principles	351
Data—the Critical Asset	352
Big Data	357
Summary	360
Chapter 23 Testing and DevOps	361
A Short Introduction to DevOps	361
DevOps and Quality	363
How Testers Add DevOps Value	371
Summary	376
Part VIII Agile Testing in Practice	379
Chapter 24 Visualize Your Testing	381
Communicating the Importance of Testing	381
Visualize for Continuous Improvement	386
Visibility into Tests and Test Results	390
Summary	392

Chapter 25 Putting It All Together	393
Confidence-Building Practices	394
Use Examples	394
Exploratory Testing	395
Feature Testing	396
Continual Learning	397
Context Sensitivity	399
Keep It Real	401
Create a Shared Vision	402
Summary	405
Appendix A Page Objects in Practice: Examples	407
An Example with Selenium 2—WebDriver	407
Using the PageFactory Class	410
Appendix B Provocation Starters	413
 Glossary	 415
References	423
Bibliography	435
Index	459

FOREWORD

By Elisabeth Hendrickson

Just ten years ago, agile was still considered radical. Fringe. Weird. The standard approach to delivering software involved phases: analyze, then design, then code, then test. Integration and testing happened only at the end of the cycle. The full development cycle took months or years.

If you have never worked in an organization with long cycles and discrete phases, the idea may seem a little weird now, but it was the standard a decade ago.

Back when phases were the norm and agile was still new, the agile community was mostly programmer-centric. Janet and Lisa and a few others from quality and testing were there. However, many in the agile community felt that QA had become irrelevant. They were wrong, of course. QA changed, reshaped to fit the new context, but it did not go away.

It took people like Janet and Lisa to show how QA could be integrated into agile teams instead of bypassed. Their first book together, *Agile Testing*, carefully explained the whole-team approach to quality. They covered the cultural changes needed to fully integrate testing with development. They explained how to overcome barriers. It's a fantastic book, and I highly recommend it.

However, questions remained. How could the practices be adapted to various contexts? How do you start? What should testers learn in order to be more effective?

This book picks up where *Agile Testing* left off and answers those questions and more.

Even if that were all this book did, it would be an excellent sequel.

It's more than that, though. Within these pages you will find a theme—one that Janet and Lisa have woven so deftly throughout the text you might not even realize it as you are reading. So I am going to call your attention to it: this is a book about adapting.

Reflect-and-adapt is the one simple trick that can enable your organization to find its way to agile. Experiment, try something different, distill lessons learned, repeat. The next thing you know, your organization will be nimble and flexible, able to shift with market demands and deliver incrementally.

This book teaches you about adapting even as it is teaching you about agile testing.

Part II, “Learning for Better Testing,” isn't just about how you learn as an individual but also about building a learning culture. Part VII, “What Is Your Context?,” isn't just about variations in agile tailored to different situations; it's also a field guide to various types of adaptations.

The world is changing so very quickly. Just a decade ago agile was weird; now it is mainstream. Just five years ago, tablets like iPads weren't even on the market; now they're everywhere. Practices, tools, technology, and markets are all changing so fast it's hard to keep up. It's not enough to learn one way of doing things; you need to know how to discover new ways. You need to adapt.

This book is a fantastic resource for agile testing. It will also help you learn to adapt and be comfortable with change.

I hope you enjoy it as much as I did.

FOREWORD

By Johanna Rothman

What do testers do? They provide information about the product under test, to expose risks for the team.

That's exactly what Janet Gregory and Lisa Crispin have done in their new book, *More Agile Testing: Learning Journeys for the Whole Team*. Do you have risks in your agility? There are plenty of ideas to help you understand the value of sustainable pace, creating a learning organization, and your role in testing.

Not sure how to test for a given product, on a single team, or in a program? There's an answer for that, too.

How do you work with people in the next cube, down the hall, and across the world? Janet and Lisa have been there and done that. Their focus on roles and not titles is particularly helpful.

There are plenty of images in this book, so you won't have to wonder, "What do they mean?" They show you, not just tell you.

More Agile Testing: Learning Journeys for the Whole Team is much more than a book about testing. It's a book about how to use testing to help your entire team, and by extension, your organization, and transition to agile in a healthy way.

Isn't that what providing information about the organization under test, exposing risks in the organization, is all about?

If you are a tester or a test manager, you need to read this book. If you integrate testing into your organization, you need to read this book. How else will you know what the testers could be doing?

This page intentionally left blank

PREFACE

Preface

Who Is This Book For?

Acceptance Tests

How to Read This Book

Experiment!

This book carries on where our first book, *Agile Testing: A Practical Guide for Testers and Agile Teams*, left off. We avoid repeating what we covered in our first book but give enough context so it stands alone if you have not read *Agile Testing*. We refer to the first book as *Agile Testing* when we think it might be helpful for the reader to explore basic concepts in more detail.

WHO IS THIS BOOK FOR?

We assume that you, the reader, are not a beginner in the world of agile testing, that you have some agile and testing experience and now you're looking for help in the areas beyond where *Agile Testing* goes. If you feel that you would like an introduction to agile development that includes some basics of testing in agile before you read this book, *The Agile Samurai* (Rasmussen, 2010) is an excellent place to start.

This book is aimed at anyone who is interested in testing activities on an agile team. In our experience, this includes not only testers and test managers, but programmers, product owners, business analysts, DevOps practitioners, line managers—pretty much everyone.

ACCEPTANCE TESTS

In addition to sharing what we've learned over the past several years, we wanted to make this book as useful to our readers as the first one. We wanted to know what readers of the first book still needed to know after

reading it, so we asked practitioners from the *Agile Testing* mailing list to send us their “acceptance tests” for this second book. We distilled those responses to this list of acceptance tests for *More Agile Testing* and did our best to satisfy these as we wrote the book.

You’ll note that we’ve used a style used in behavior-driven development (BDD), which we’ll talk more about in Chapter 11, “Getting Examples”:

Given <precondition>,
When <trigger, action>,
Then <the expected result>.

- Given that I am an agile tester or manager, when I hire new testers with no agile experience, then I’ll learn how to bring them up to speed and avoid throwing them into the deep end without a life jacket.
- Given that I am a team member on an agile team, when I finish this book, then I expect to know how to fit exploratory testing in with automated tests and to get a picture of the overall test coverage, without resorting to heavyweight tools.
- Given that I am an experienced agile test manager, when I finish this book, then I will understand how to approach agile testing techniques with multiple teams to allow my successful agile organization to grow.
- Given that I am an experienced agile test manager, when I finish reading this book, then I should have ideas about how to coordinate test automation activities across iterations and teams, with ideas on how to improve.
- Given that I am an experienced agile manager, when I’ve read this book, then I will understand how other teams have adapted agile testing practices to suit their own context and will have ideas about how to apply them to mine.
- Given that I am an agile team member who is interested in testing, when I finish this book, then I expect to have examples of what tests should and should not look like and how I can design tests effectively.

- Given that I am an experienced agile tester, when I find an interesting topic in this book about which I'd like to learn more, then I can easily find references to web resources or other books.
- Given that I am an experienced agile coach or manager who is reading the book, when I see a concept that would help my team, then I have enough information to be able to devise a strategy to get the team to try an experiment.
- Given that I am an agile team member who is concerned about testing and keeping the customers informed, when I have read this book, then I'll understand good ways to communicate with customer team members about testing activities.
- Given that I am an experienced agile test manager, when I have read this book, then I will know how mainstream adoption of agile is being done, and I will understand the working context of testers from other organizations when they apply for jobs on my team. (*Note:* This acceptance test is not part of this release, but we think some of the examples and stories in the book will help to achieve it.)

HOW TO READ THIS BOOK

Though we've organized this book in a way that we feel flows best, you don't have to start with Chapter 1 and keep going. As with *Agile Testing*, you can begin with whatever topics are most useful to you. We try to cover each topic in detail only once, but because so many of these concepts, practices, and principles are interrelated, you'll find that we refer to some ideas in more than one chapter.

Part I: Introduction

Read this part to understand where testing started in agile teams and how it has evolved to become the cornerstone of agile development and continuous delivery of products. Part of successful agile development is an organization's ability to learn what's most critical for long-range success with agile testing.

- Chapter 1, "How Agile Testing Has Evolved"
- Chapter 2, "The Importance of Organizational Culture"

Part II: Learning for Better Testing

Both technology and the craft of testing are continually evolving, and lines between different disciplines are becoming more blurred. Even experienced practitioners have to keep growing their skills. This part includes examples of what testers and other disciplines such as business analysis and coding need to know to meet more difficult testing challenges. We explain the benefits of generalizing specialists and list some of the intangible thinking skills and specific technical testing skills that help testers and teams improve. Different aspects of what and how to learn are covered in the following chapters:

- Chapter 3, “Roles and Competencies”
- Chapter 4, “Thinking Skills for Testing”
- Chapter 5, “Technical Awareness”
- Chapter 6, “How to Learn”

Part III: Planning—So You Don’t Forget the Big Picture

Planning “just enough” is a balancing act. While we need to work in small increments, we have to keep an eye on the larger feature set and the entire system. This part covers different aspects of test planning, from the release level down to the task level. It also explores different models such as the agile testing quadrants and some of the adaptations people have suggested.

- Chapter 7, “Levels of Precision for Planning”
- Chapter 8, “Using Models to Help Plan”

Part IV: Testing Business Value

If, like so many agile teams, you deliver robust code in a timely manner, only to find it isn’t what the customers wanted after all, the information in this part will help. We cover tools and practices, particularly those from the agile business analysis profession, to help you test ideas and assumptions early and ensure that everyone knows what to deliver. We

address other overlapping disciplines and expanding mindsets. This is a big area, so there are several chapters:

- Chapter 9, “Are We Building the Right Thing?”
- Chapter 10, “The Expanding Tester’s Mindset: Is This *My* Job?”
- Chapter 11, “Getting Examples”

Part V: Investigative Testing

The programmers have delivered some code to test. Where do you start? If you or your team lacks experience with exploratory testing, you’ll find some help here. We outline several exploratory testing techniques such as using personas and tours to help generate test charter ideas, as well as managing charters with session-based test management and thread-based test management.

Along with all those different ways to do exploratory testing, we look at other ways to verify that delivered code meets a wide range of business and user needs. This part covers ways to mitigate risks and generate useful information in several different types of testing that present challenges to agile teams. The investigative testing chapters are

- Chapter 12, “Exploratory Testing”
- Chapter 13, “Other Types of Testing”

Part VI: Test Automation

We see more and more teams finding ways to succeed with test automation. However, for many teams, automated tests produce sporadic failures that are expensive to investigate. The time (cost) spent on each failure may be more than the test is worth. There are plenty of pitfalls in automating tests. In this part we give examples of ways to make technical debt in testing visible. We look at different ways to use the agile testing pyramid effectively to help you think about how to plan your automation. We’ve introduced a few alternative pyramid models

to approach automation from different perspectives. You'll learn ways to design automated tests for optimum reliability and ease of maintenance. This part also includes examples of scaling test automation in a large enterprise company.

The chapters in Part VI are

- Chapter 14, “Technical Debt in Testing”
- Chapter 15, “Pyramids of Automation”
- Chapter 16, “Test Automation Design Patterns and Approaches”
- Chapter 17, “Selecting Test Automation Solutions”

Part VII: What Is Your Context?

Your approach to agile testing will naturally depend on your context. Do you work with large enterprise systems? Maybe you're newly tasked with testing mobile apps or embedded software. Perhaps your team is challenged with finding good ways to test data that helps businesses make decisions. Have you wondered how agile can work in testing regulated software? Finally, we look at the synergies between testing and the DevOps movement. The chapters in this part cover a variety of areas, so we have included a number of stories from people who are currently working in those situations. Some of these chapters may not apply to your working environment today, but tomorrow—who knows?

- Chapter 18, “Agile Testing in the Enterprise”
- Chapter 19, “Agile Testing on Distributed Teams”
- Chapter 20, “Agile Testing for Mobile and Embedded Systems”
- Chapter 21, “Agile Testing in Regulated Environments”
- Chapter 22, “Agile Testing for Data Warehouses and Business Intelligence Systems”
- Chapter 23, “Testing and DevOps”

Part VIII: Agile Testing in Practice

We wrap up the book with a look at how teams can visualize quality and testing, and a summary of agile testing practices that will give your team

confidence as you make release decisions. Creating a shared vision for your team is critical to success, and we share a model to help bring testing activities to the whole team. If you're feeling a bit overwhelmed right now and aren't sure where to start, read these chapters first:

- Chapter 24, "Visualize Your Testing"
- Chapter 25, "Putting It All Together"

The book also includes two appendixes: Appendix A, "Page Objects in Practice: Examples," and Appendix B, "Provocation Starters."

Other Elements

Since teams use such a wide variety of agile practices and approaches, we've tried to keep our terminology as generic as possible. To make sure we have a common language with you, we've included a glossary of the terms we use.

You'll find icons in the margins throughout the book where we'd like to draw your attention to a specific practice. You'll find all six icons in Chapter 1, "How Agile Testing Has Evolved," and Chapter 25, "Putting It All Together." An example of the icon for learning can be seen next to following paragraph.



We hope you'll want to learn more about some of the practices, techniques, and tools that we cover. Please check the bibliography for references to books, websites, articles, and blogs. We've sorted it by part so you can find more information easily when you're reading. Sources that are mentioned directly in the book are listed alphabetically in the reference list for easy lookup.

The mind map overview from *Agile Testing* is included on the book website, www.agiletester.com, so that you can get a feel for what was covered there if you haven't already read it.

EXPERIMENT!

Linda Rising encouraged us years ago to try small experiments, evaluate the results, and keep iterating to chip away at problems and achieve goals. If you read something in this book that sounds as if it might be useful for you or your team, give it a try for an iteration or two. Use your retrospectives to see if it's helping, and tweak as necessary. If it doesn't work, you learned something, and you can try something different.

We hope you will find many experiments to try in these pages.

ACKNOWLEDGMENTS

This book has been a group effort. Please learn about all the wonderful practitioners who shared their stories as sidebars in “About the Contributors.” Many are success stories, some describe lessons learned the hard way, but we know all will benefit you, the reader.

We’re extremely grateful to Jennifer Sinclair for her wonderful illustrations. She came up with such creative ideas to help us get across some important concepts.

We referenced the ideas of so many other people who have taken ideas from *Agile Testing*, adapted them to meet their needs, and were willing to share with the world—thank you.

Our tireless reviewers helped us shape the book and cover the right topics. We’re especially grateful to Mike Talks, Bernice Niel Ruhland, and Sherry Heinze, who slogged through every chapter, in some cases multiple times. Thanks to Augusto Evangelisti, Gojko Adzic, Adam Knight, Steve Rogalsky, Aldo Rall, Sharon Robson, James Lyndsay, JeanAnn Harrison, Ken Rubin, Geoff Meyer, Adam Yuret, and Mike Cohn for their valuable feedback. Each of our story contributors also helped review the chapters that included their stories.

Special thanks to our technical reviewers, whose feedback on our final draft was immensely helpful: Tom Poppendieck, Liz Keogh, Markus Gärtner, and George Dinwiddie.

Thank you, Christopher Guzikowski, for making this book possible in the first place, and Olivia Basegio, for answering a thousand questions and keeping us organized. We are grateful to our developmental